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**Project:** AD-Fusion-v2

**Subject:** Pipeline Standardization and DevOps Procedures

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## 1. Introduction and Objectives

The purpose of this documentation is to codify the environment standards for the **AD-Fusion-v2** project. To ensure project continuity and eliminate the "it works on my machine" variable, we are transitioning all development and training to a unified Docker-based architecture. This log details the resolution of recent numerical instabilities and establishes the hardware constraints required for data privacy compliance.

## 2. Technical Stack Requirements

To leverage the full capabilities of our local **H100 cluster**, the following dependency versions are now mandatory. Any deviation from these versions must be approved by me to prevent breakage in the downstream synthesis pipeline.

- **Base Image:** Ubuntu 22.04 LTS
- **CUDA Toolkit:** 12.1
- **Python:** 3.10.12
- **Frameworks:** \* **PyTorch:** 2.4.0
  - **MONAI:** 1.3.0

## 3. Numerical Instability Resolution (NumPy/SciPy Conflict)

During the week of Jan 11th, **Bob** identified a critical failure in the data preprocessing pipeline where several MRI volumes were returning \$NaN\$ (Not a Number) values during spatial normalization.

### 3.1 Root Cause Analysis

Upon auditing Bob's local environment, I discovered a version mismatch between **NumPy** and **SciPy**. The local environment was running an outdated version of **SciPy** (1.9.x) against **NumPy** 1.26. This caused a regression in the underlying C-API calls during heavy matrix operations, specifically during the calculation of affine transformations. This resulted in floating-point overflows, producing the \$NaN\$ values that halted Bob's progress.

### 3.2 Standardized Solution

I have resolved this by creating a custom **Dockerfile** that enforces strict version pinning. By utilizing an **Ubuntu 22.04** base, we ensure that the system-level linear algebra libraries (BLAS/LAPACK) are consistent.

**Action for Bob:** Please deprecate your local Conda environment immediately. All further preprocessing must occur within the containerized environment to ensure mathematical reproducibility.

## 4. Security and Compliance: Local H100 Enforcement

**Alice** has provided a formal update regarding our data handling. Due to the sensitivity of the AD-Fusion datasets, **cloud processing (AWS, GCP, or Azure) is strictly prohibited** until the Data Processing Agreement (DPA) has been signed by all stakeholders.

- **Training Constraint:** All training runs must remain on our local **H100** cluster.
- **Task for Chloe:** Please update the training scripts to point to our internal `/datasets/AD-Fusion-v2/` path. Ensure that no data is being cached to external cloud-synced directories (e.g., Dropbox or OneDrive).
- **Task for Bob:** Ensure your preprocessing outputs are written directly to the secure internal NAS.

## 5. How-To: Accessing the Standardized Environment

The latest environment image has been pushed to our internal GitLab Container Registry. Follow these steps to synchronize your workspace.

### 5.1 Authentication

First, authenticate with the internal registry:

Bash

```
None
docker login gitlab.internal.research:5050
```

### 5.2 Pulling and Running

Use the following commands to pull the image and launch your development container with GPU access:

Bash

```
None
# Pull the latest image
docker pull
gitlab.internal.research:5050/ad-fusion/v2-env:latest

# Run the container with H100 GPU support
docker run --gpus all -it \
  -v /home/user/project:/workspace \
  gitlab.internal.research:5050/ad-fusion/v2-env:latest
/bin/bash
```

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## 6. Action Items and Assignments

| Task                                                                                | Owner         | Priority | Due Date   |
|-------------------------------------------------------------------------------------|---------------|----------|------------|
| <b>Verify NaN Fix:</b> Test the new container on the T1-weighted batch.             | <b>Bob</b>    | High     | 2026-01-20 |
| <b>Pipeline Integration:</b> Integrate MONAI 1.3.0 transforms into the main branch. | <b>Chloe</b>  | Medium   | 2026-01-22 |
| <b>DPA Follow-up:</b> Provide an update on the legal status of cloud processing.    | <b>Alice</b>  | High     | 2026-01-25 |
| <b>Registry Maintenance:</b> Prune old development images to save NAS space.        | <b>Daniel</b> | Low      | 2026-01-21 |

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